

1	(a)	$W = \text{J s}^{-1} = \text{kg m}^2 \text{s}^{-3}$ SI base units of $p = \frac{\text{kg m}^2 \text{s}^{-3}}{\text{m K}}$ $= \text{kg m s}^{-3} \text{K}^{-1}$		C1 A1
	(b)	(i)	$q = py^2 \left(\frac{T}{x} \right)$ $= (237)(0.0020)^2 \left(\frac{323}{1.0} \right)$ $= 0.306$	C1 A1
		(ii)	$\frac{\Delta q}{q} \times 100\% = \left(\frac{\Delta p}{p} + 2 \frac{\Delta y}{y} + \frac{\Delta T}{T} + \frac{\Delta x}{x} \right) \times 100\%$ $= 2.0 + (2 \times 3.0) + 0.5 + 1.0$ $= 9.5\%$	C1 A1
	(c)	$\Delta q = \frac{9.5}{100} (0.306)$ $= 0.03$ $q = 0.31 \pm 0.03$ (actual uncertainty to 1 s.f as well as dp of q and actual uncertainty must match)		C1 A1

2	(a)	-9.81 m s^{-2}	
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